

Project Documentation

IMDb Movie Data Analysis & Visualization

1. Project Overview

This project analyses a comprehensive dataset of over 5,000 movies using Microsoft Excel and Power BI. The objective is to uncover trends in box office revenue, evaluate director performance, and assess global market profitability to improve data-driven decision-making in the entertainment sector.

2. Tools Used

- **Microsoft Excel:** Used for data cleaning, handling missing values, and formatting.
- **Microsoft Power BI:** Used for data modelling, DAX measure creation, and interactive dashboard visualization.

3. Dataset

- **Source:** IMDb Movie Dataset (movie_metadata-DATASET.csv)
- **Data Contains:** 28 columns covering key metrics such as Movie Title, Director Name, Gross Revenue, Budget, Genres, Content Rating, Country, Language, and IMDb Score.

4. Steps Followed

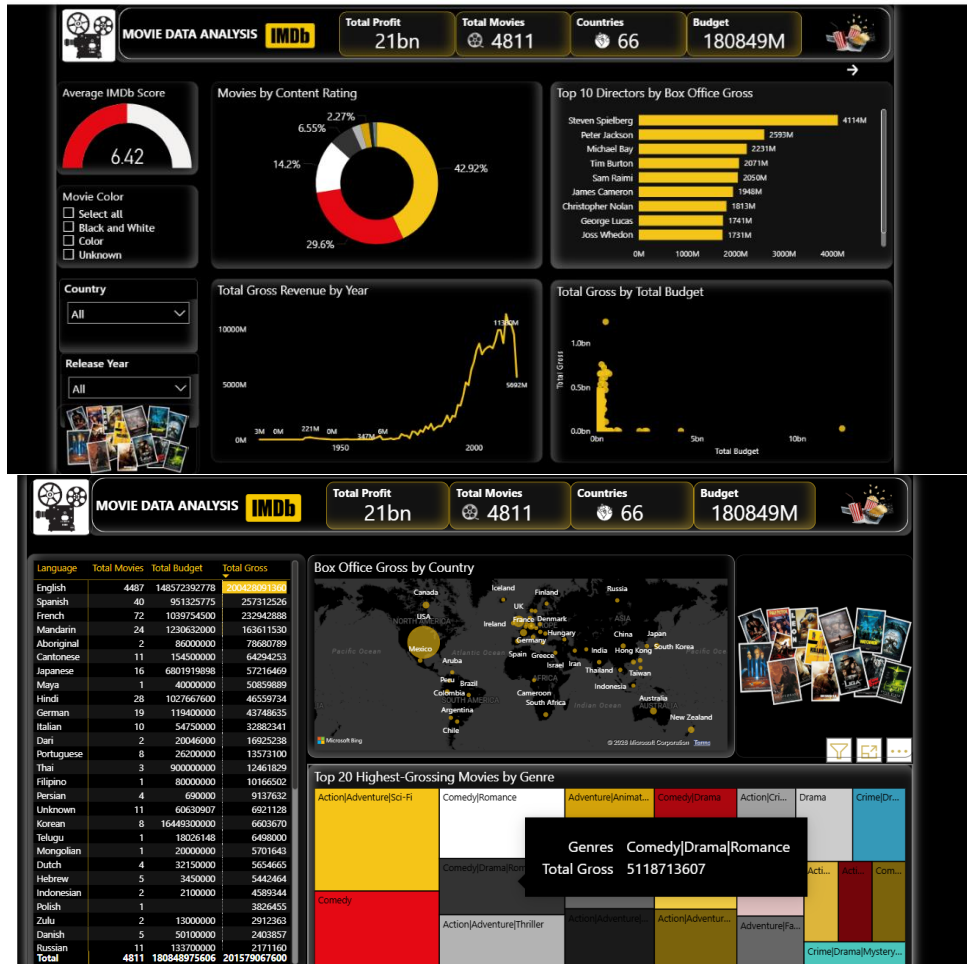
- **Cleaned data in Excel:** Removed duplicates, handled missing values (e.g., replaced blank years with a 1900 placeholder), standardized text, and formatted financial columns.
- **Imported cleaned data into Power BI:** Established data types and prepared the dataset for modelling.
- **Created custom DAX measures:** Calculated key metrics like Total Gross, Total Budget, Total Profit, and Total Movies.
- **Built dashboards using charts, slicers, and KPIs:** Designed a cinematic, dark-themed interactive dashboard featuring Map visuals, Tree maps, Donut charts, Line charts, and a dedicated Director Drill-through page.

5. Key Insights

- **Geographic Dominance:** The USA market accounts for the vast majority of global box office revenue compared to international markets.
- **Top Performing Genres:** Action, Adventure, and Sci-Fi are the highest-grossing genres and contain the most top-performing blockbuster titles.
- **Director Impact:** A select group of directors (e.g., Steven Spielberg, Peter Jackson) consistently generate the highest box office returns.

- **Content Demographics:** R and PG-13 are the most frequently produced content ratings, indicating the primary target demographic for major studio releases.

6. Screenshots



7. Files Included

- movie_metadata -CLEANED.xlsx – Cleaned dataset used for analysis.
- movie_metadata-DATASET.csv – The original raw dataset.
- Movie Metadata-DASHBOARD.pbix – The interactive Power BI dashboard file.
- README.md – GitHub repository description and project summary.

8. How to Use

- Open movie_metadata -CLEANED.xlsx to view the finalized, cleaned data table.
- Open Movie Metadata-DASHBOARD.pbix in Power BI Desktop to explore the visuals, use the genre/year slicers, and test the drill-through features.